An *Ab Initio* Rovibrational Linelist for HPO

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Motivation: Why HPO?

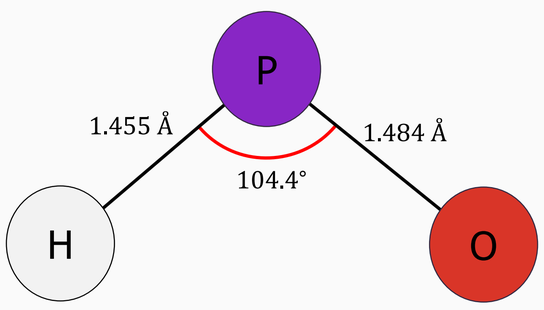
HPO appears as a precursor molecule in proposed abiotic mechanisms to form phosphine on Venus[1]. It has been used for modeling the atmosphere of 55 Cancri e and predicted to be detectable in the interstellar medium[2,3].

Can HPO play a role on lava or rocky exoplanets?

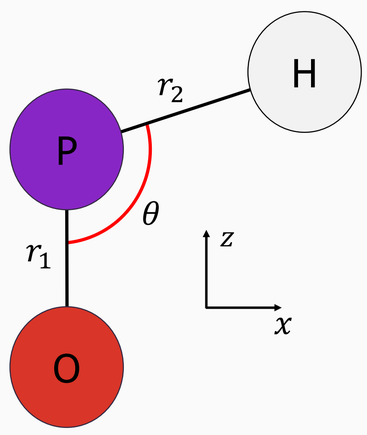
Aims

- Produce a rovibrational linelist for HPO (electronic ground state only).
- Evaluate potential for HPO detection on exoplanets/planets.

Equilibrium bond lengths and angles at the electronic ground state ($\tilde{X}\ ^1A'$)

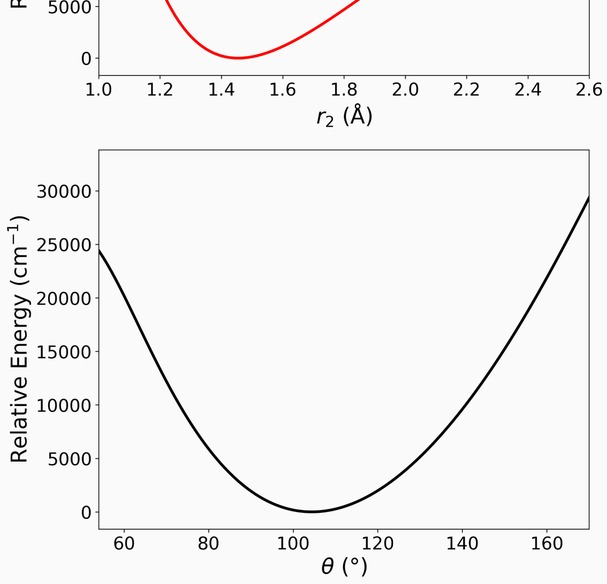


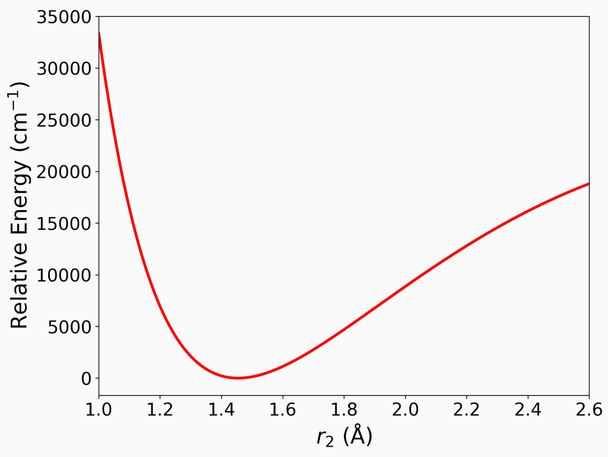
We use the internal coordinates and Cartesian axis as presented:



Potential Energy Surface

7770 *ab initio* points computed with CCSD(T)-F12/cc-pVQZ-F12a

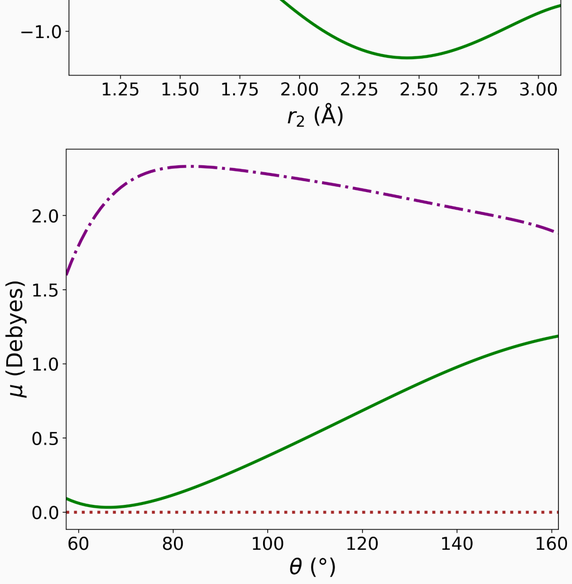


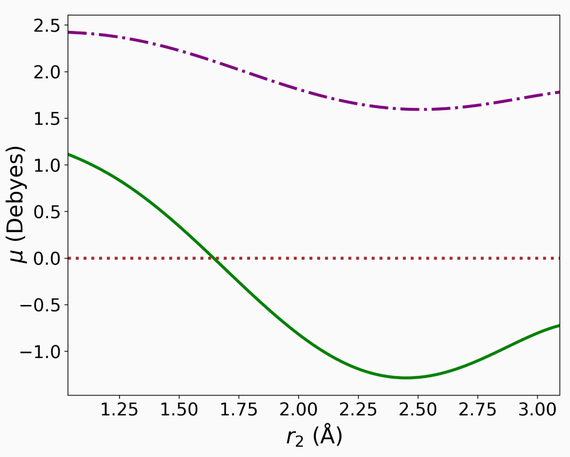


- Fit PES has root-mean-square error of 2 cm^{-1} for energies under 18,000 cm^{-1} .
- Molecule appears relatively rigid (stiff bending), we can use non-singular methods to perform variational calculations.

Dipole Moment Surfaces

Same grid as the PES computed with CCSD(T)/aug-cc-pV(5+d)Z

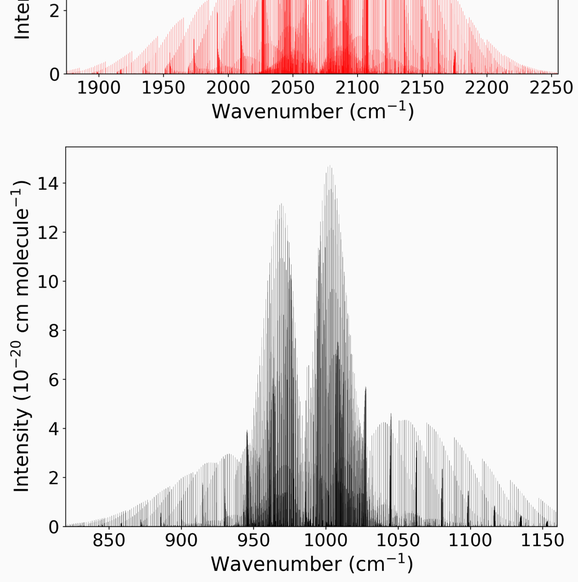


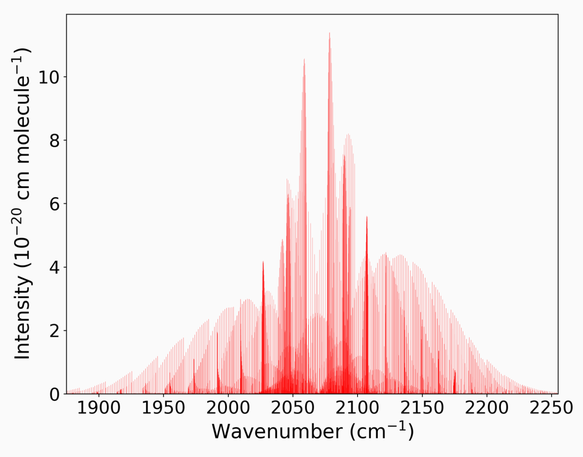


- Dipole values calculated oriented with respect to the Cartesian axis given.
- Fitted to a Taylor-series type expansion. Dipole values along the y-axis are 0 since the molecule is planar.

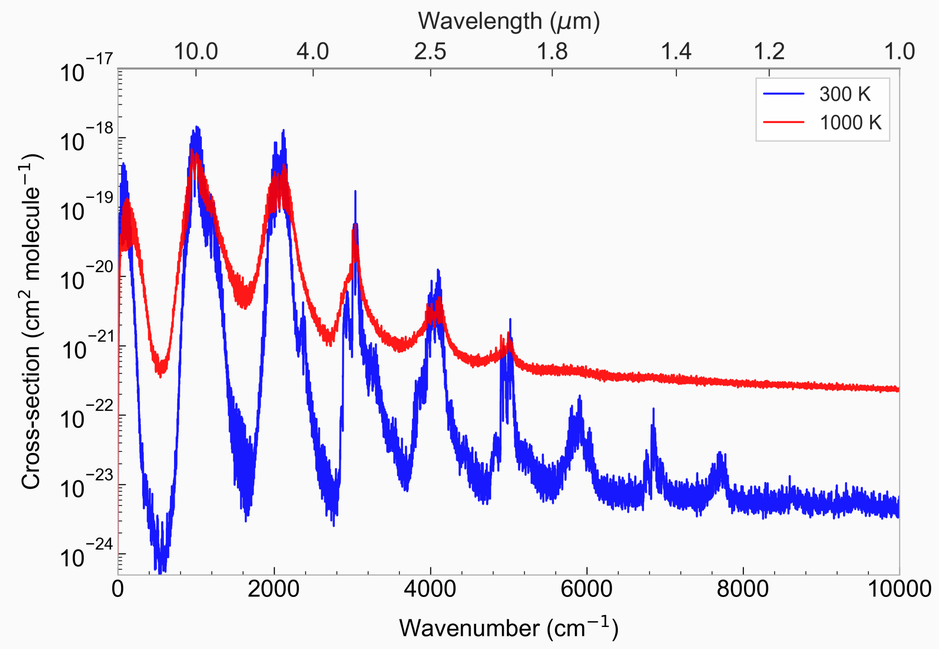
Stick Spectra

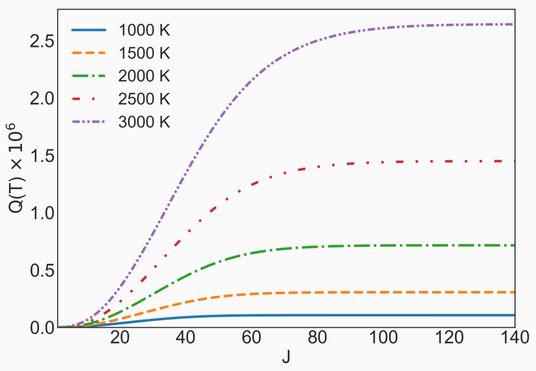
Rotational stick spectra of the fundamental vibrational modes





- The first and third fundamental modes have been experimentally measured, but not the second, despite intensity seeming higher.
- Experimental values can be used to refine rovibrational calculations.





Cross-sections computed with states up to J = 141. Partition function plots as a function of J suggests J = 120 could be sufficient instead.

Cross-sections and partition functions were computed with the program ExoCross[5].

Cross-sections

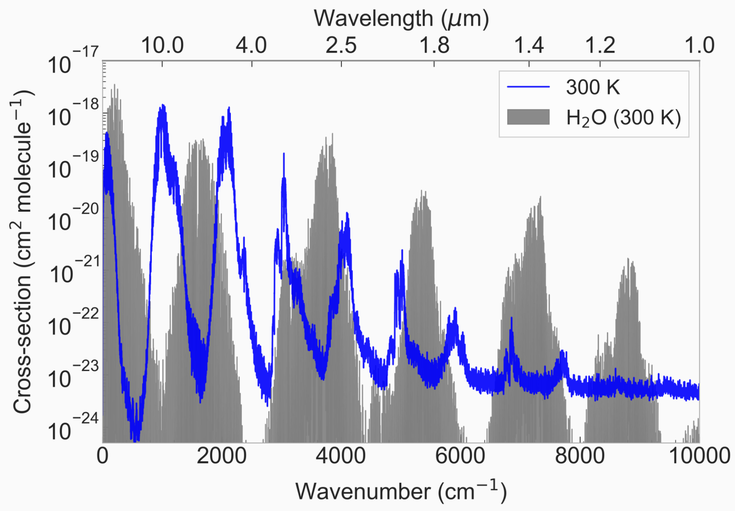
Rovibrational energies and wavefunctions are computed with variational nuclear motion program TROVE[4], then used to compute stick spectra and cross sections.

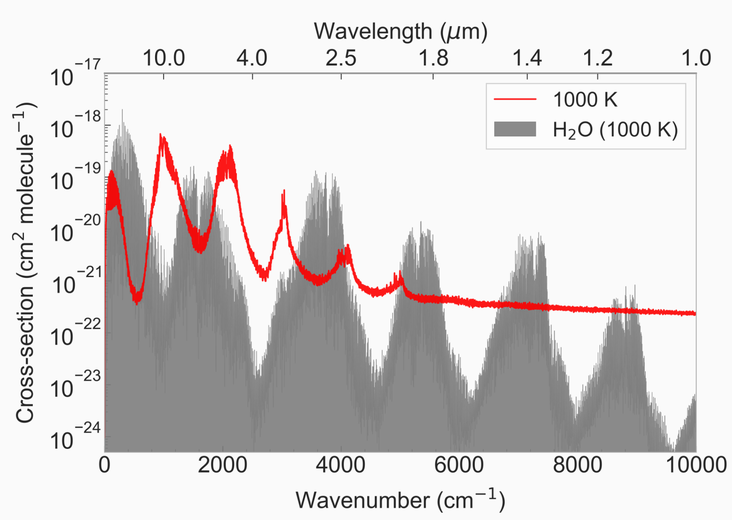
Lower states are selected between 0 to 14,000 cm^{-1} while upper states are between 0 to 24,000 cm^{-1}. Includes 2.4 million rovibrational states and 2.2 x 10^{10} transitions.

Exoplanet Detection?

The fundamental bands of HPO do not overlap with water. This makes detection more feasible. The fundamental band at 985 cm^{-1} would be particularly visible.

We suggest HPO can be of interest for studying exoplanets and the planetary chemistry of phosphorus. A line list for HPO should be provided on the ExoMol website soon.





References:
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4. S. N. Yurchenko, W. Thiel, and P. Jensen, J. Mol. Spectrosc. 245, 126 (2007).
5. S. N. Yurchenko, A. F. Al-Refaie and J. Tennyson, 'EXOCROSS: a general program for generating spectra from molecular line lists', A&A, 614 (2018) A131 DOI: https://doi.org/10.1051/0004-6361/201732531

ExoMol

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